



INSTITUT TEKNOLOGI DEL

MATERI PRAKTIKUM

4233201 – Kecerdasan Buatan

Semester Ganjil Tahun Ajaran 2025/2026

Session Date	: 08 September 2025
Week/Session	: 3/2
Topic	: Simple Linear Regression & Multiple Linear Regression
Practicum Goals	: Mahasiswa mampu melatih model regresi, menentukan persamaan garis regresi dan mengevaluasi performa model
Duration	: 2 x 50 minutes
Place of Submission	: ecourse
Submission Deadline	: 08 September 2025, jam 23.50

Simple Linear Regression

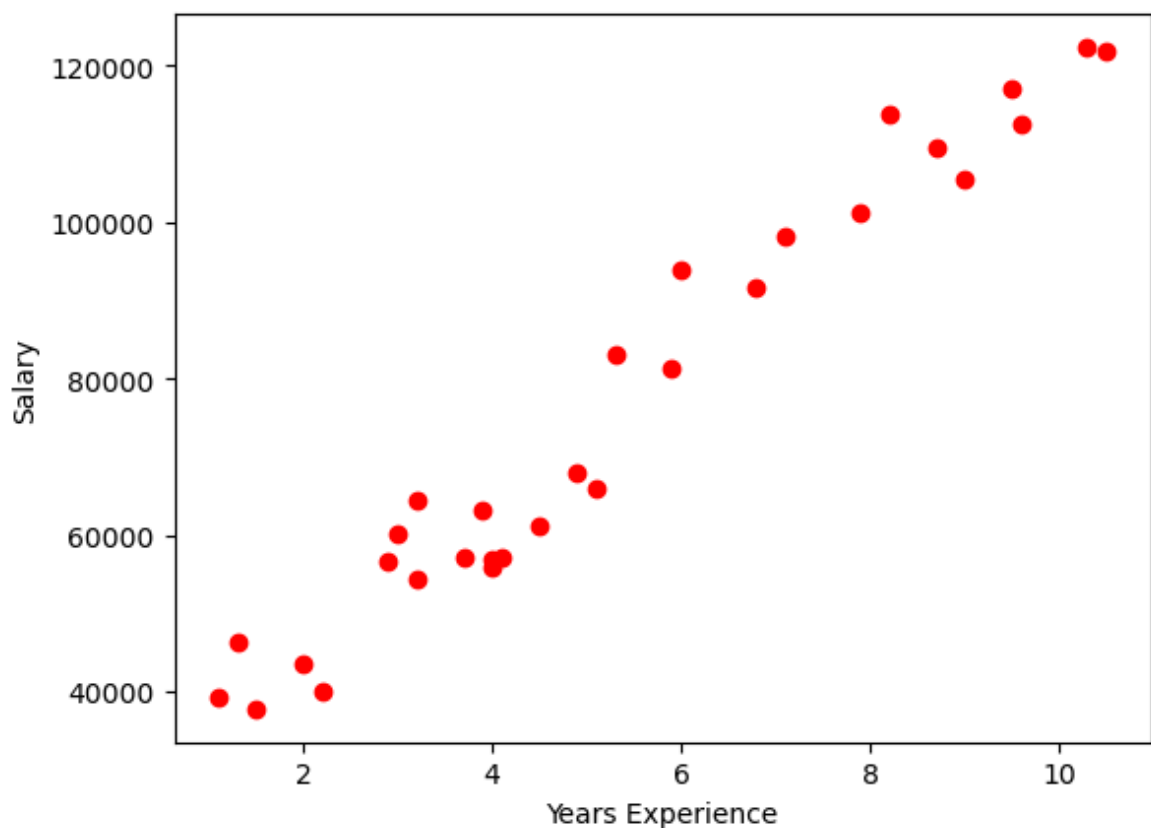
```
In [1]: import pandas as pd
df = pd.read_csv('Salary_Data.csv')
print(df.head())
```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

Menentukan predictor (X) dan target (y) dari DataFrame untuk Simple Linear Regression

```
In [2]: X = df.iloc[:, :-1].values
y = df.iloc[:, 1].values
```

```
In [3]: import matplotlib.pyplot as plt
plt.scatter(X, y, color = "red")
plt.xlabel('Years Experience')
plt.ylabel('Salary')
plt.show()
```



```
In [4]: #bagi dataset jadi train dan test 80:20
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

Melatih model regresi linear sederhana pada data X_{train} (variabel independen) dan y_{train} (target) menggunakan fungsi `LinearRegression()`. Setelah model dilatih, ia akan

menyesuaikan garis regresi terbaik untuk memprediksi nilai target berdasarkan variabel input

```
In [5]: # Fitting Simple Linear Regression to the Training set
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train, y_train)
```

```
Out[5]: ▼ LinearRegression ⓘ ?
LinearRegression()
```

- Menghitung nilai *intercept* a , yaitu nilai y saat $x = 0$ dalam persamaan $\hat{y} = a + bx$
- Menghitung *slope* atau koefisien kemiringan b yang menunjukkan seberapa besar perubahan y saat x bertambah

```
In [6]: a = regressor.intercept_
b = regressor.coef_[0]

print("Estimated model intercept, a:", a)
print("Estimated model slope, b:", b)
print(f"ŷ = {a:.2f} + {b:.2f}x")
```

Estimated model intercept, a: 25321.583011776813

Estimated model slope, b: 9423.815323030976

$\hat{y} = 25321.58 + 9423.82x$

```
In [7]: # memprediksi hasil pada data uji (Test set) menggunakan model regresi yang telah
y_pred = regressor.predict(X_test)
y_pred
```

```
Out[7]: array([115790.21011287,  71498.27809463, 102596.86866063,  75267.80422384,
        55477.79204548,  60189.69970699])
```

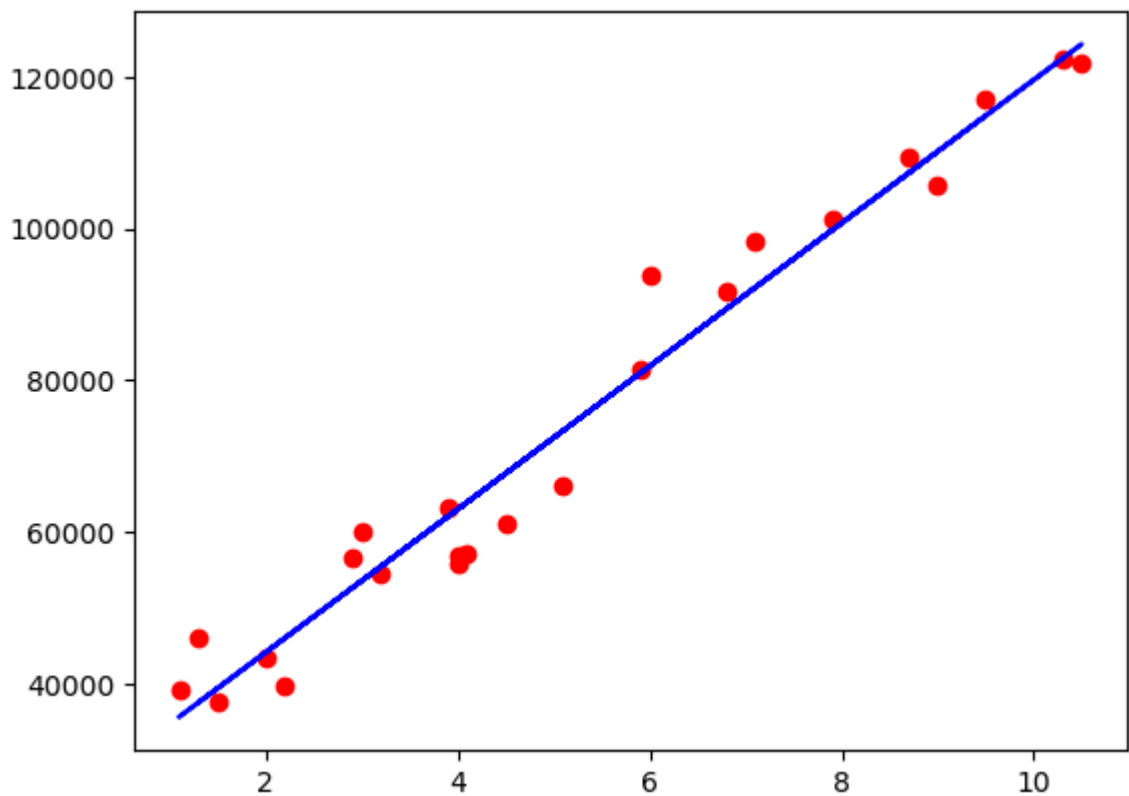
```
In [8]: # Menghitung nilai R-Square, yang mengukur seberapa baik model regresi linear
# Semakin mendekati 1, semakin baik model
```

```
from sklearn.metrics import r2_score
score = r2_score(y_test, y_pred)
score
```

```
Out[8]: 0.9024461774180497
```

```
In [9]: # Visualizing the Training set results
plt.scatter(X_train, y_train, color = 'red')
plt.plot(X_train, regressor.predict(X_train), color = 'blue')
```

```
Out[9]: [<matplotlib.lines.Line2D at 0x1eb999c0500>]
```



```
In [10]: # Visualizing the Test set results
plt.scatter(X_test, y_test, color = 'red')
plt.plot(X_train, regressor.predict(X_train), color = 'blue')
plt.title('Salary vs Experience (Test set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()
```



Multiple Linear Regression

```
In [11]: df2 = pd.read_csv('multiple_linear_regression_dataset.csv')
print(df2.head())
```

	age	experience	income
0	25	1	30450
1	30	3	35670
2	47	2	31580
3	32	5	40130
4	43	10	47830

Menentukan predictor (X) dan target (y) dari DataFrame untuk Multiple Linear Regression

```
In [12]: X2 = df2.iloc[:, :-1]
y2 = df2.iloc[:, 2]
```

```
In [13]: # Bagi dataset menjadi train dan test 80:20
```

```
X_train2, X_test2, y_train2, y_test2 = train_test_split(X2, y2, test_size = 0.2,
```

Multiple Linear Regression juga menggunakan fungsi yang sama seperti Simple Linear Regression, yaitu `LinearRegression()` dari Scikit-Learn.

```
In [14]: regressor = LinearRegression()
regressor.fit(X_train2, y_train2)
```

```
Out[14]: LinearRegression
```

```
LinearRegression()
```

```
In [15]: # Predicting the Test set results
y_pred2 = regressor.predict(X_test2)
print(y_pred2)
```

```
[31093.38107376 31295.49954076 40250.46080162 34897.6958918 ]
```

- Menghitung nilai *intercept* a , yaitu nilai y saat $x = 0$ dalam persamaan $\hat{y} = a + bx$
- Menghitung *slope* atau koefisien kemiringan b yang menunjukkan seberapa besar perubahan y saat x bertambah
- Karena pada Multiple Linear Regression ada lebih dari satu predictor, maka koefisien ditentukan sesuai dengan jumlah predictor

```
In [16]: a_ = regressor.intercept_
b1 = regressor.coef_[0]
b2 = regressor.coef_[1]

print("Estimated model intercept, a:", a_)
print("Estimated model slope, b1:", b1)
print("Estimated model slope, b2:", b2)
print(f"ŷ = {a_:.2f} {b1:.2f}x1 + {b2:.2f}x2")
```

Estimated model intercept, a: 31465.056418503937
Estimated model slope, b1: -101.05923350088756
Estimated model slope, b2: 2154.805492774648
 $\hat{y} = 25321.58 - 101.06x_1 + 2154.81x_2$

```
In [17]: # Menghitung nilai R-Square, yang mengukur seberapa baik model regresi Linear  
# Semakin mendekati 1, semakin baik model  
  
score2=r2_score(y_test2,y_pred2)  
score2
```

Out[17]: 0.9387098237077887

Tugas

- Simple Linear Regression: prediksi score (y) berdasarkan waktu belajar (X) dari dataset <https://www.kaggle.com/datasets/snmahsa/student-score-suitable-for-linear-regression>
- Multiple Linear Regression: prediksi sales (y) berdasarkan data marketing (x) di youtube, facebook, newspaper, dari dataset <https://www.kaggle.com/datasets/fajejavad/marketing-linear-multiple-regression>